The cross-linguistic prevalence of the sonority sequencing principle in adult and child-directed speech

Introduction

Speech is made of hierarchically organised, discrete linguistic units (e.g., phonemes, syllables, words, phrases). During early language acquisition, syllables are a privileged linguistic unit in speech processing: newborns preferentially chunk the speech signal into syllable-sized units (Bijeljac-Babic et al., 1993; Fló et al., 2019). This is remarkable, given that they have accumulated relatively limited linguistic experience. To date, how newborns are able to identify syllables in speech is unclear. Previous work points to newborns being sensitive to certain constraints to syllabic structure. One of them is the Sonority Sequencing Principle (SSP).

The SSP constrains the order of the segments inside a syllable based on sonority, a phonological feature that describes the perceptual salience of a segment relative to neighbouring ones. Although the specific acoustic correlates of sonority are still under debate (Albert & Nicenboim, 2022; Parker, 2008), sonority scales have been developed and agreed one by most researchers so that different segment classes are assigned some scalar, ordered sonority score, following available literature of cross-linguistic phonology (Jany et al., 2007, for an acoustic grouding of this sonority hierarchy, based on intensity). By assigning numeric sonority values to specific segments, syllables can be classified as following or violating the SSP (Blevins, 1995; Clements, 1990)

The SSP states that each syllable displays a unique peak of sonority in the nucleus, and that sonority rises monotonically from the syllable margins toward the nucleus (Parker, 2008). Gomez et al. (2014) showed in newborns that hearing SSP-following syllables (e.g., *bran*) elicits different brain haemodynamic responses compared to SSP-violating syllables (e.g., *lbif*), suggesting that newborns, how have not yet acquired extensive experience with the phonological contraints of their native language, are already sensitive to violations of the SSP. Such early bias toward certain syllabic structures would provide humans with an available mechanism to detect syllables in the continuous speech stream at the earliest stages of language acquisition.

The SSP has been long claimed to be a linguistic universal, on the grounds of SSP-following syllables being preferred over SSP-violating syllables across most languages. For instance, Parker (2017) collected evidence from 275 experimental works on a typologically diverse list of 75 distinct languages, and found that 63% of the studies suggest that sonority plays a relevant role for accounting for the results, and 22% provide inconclusive evidence, and 15% provide negative evidence for such a relevant role of sonority. The universality of the SSP would place this principle as one of the few cues for speech segmentation present from birth to *all* humans, before language-specific constraints to sound sequences (e.g., phonotactics) are available. The universal status of the SSP is however a matter of debate, as violations of the SSP are common across

languages (Yin et al., 2023). For instance, onset consonant clusters of the type initial fricative-stop consonant (e.g., /st/ as in *Stein* in German) or its inverse counterpart in coda stop consonant-fricative (e.g., /ks/ as in *clicks* in English) are attested in many languages (Kawasaki-Fukumori et al., 1997). Some languages, like Russian, are known for allowing fairly complex consonant clusters at syllabic onset that violate the SSP (e.g., the monosyllabic /vzglyad/ (взгляд [glance]) (Halle & Jones, 1971; Itô, 1982). More recent accounts of linguistic universality (Prince & Smolensky, 2004) propose a softer understanding of the term "universal", such that linguistic universals represent overall preferences toward certain linguistic structures across languages, instead of hard ON/OFF constraints. Under this account, the role of the SSP as a universally available cue for syllabification becomes a quantitative question: how often do syllables in different languages abide by the SSP? (Frisch, 2015, for a similar approach).

The aim of the present paper is to provide empirical estimates of the prevalence of the SSP in a fairly representative sample of 139 languages from 17 distinct language families. We pursued this goal in two ways. In a first analysis, we estimated the proportion of the syllabic repertoire of each language that followed SSP. Using Bayesian modelling we derived a quantitative estimate of global and language-specific SSP prevalence, along with their corresponding uncertainty. Because items inside the syllabic repertoire of a language likely occur at different rates in speech (e.g., both /ka/ and /gla/ syllables are attested in Spanish, but the former likely appears in a larger number of words), a second analysis investigated whether SSP-following syllables are also more frequent speech compared to SSP-violating syllables. We explored this research question in two datasets. In the first dataset, syllabic frequencies were derived from available databases of lexical frequencies that did not discriminate between adult-directed speech (ADS) and child-directed speech (CDS). In the second dataset, syllabic frecuencies were derived from lexical frequencies in the CHILDES (MacWhinney & Snow, 1990), which includes corpora of child-directed speech.

Analysis 1: What proportion of the syllabic repertoire of a language follows the SSP?

Methods

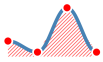
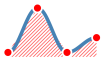
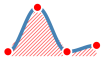
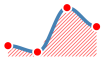
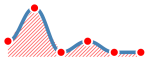
Data collection

We scraped all phonological transcriptions from word entries in the Wiktionary database. Wiktionary is an open-source, free database of lexical entries. As of June 10th 2026, Wiktionary contains 10,050,751 lexical entries from 8,244 langages contributed by 4,463,976 total users. Some lexical entries in Wiktionary also include a phonological in IPA format or phonetic transcription of the word-form in a particular language. We used the `wikipron` Python package (Lee et al., 2020) to scrape all available phonological transcriptions from all 7859 languages included in the Glottolog database (Nordhoff & Hammarström, 2011). This resulted in ~1.7 million pronunciations belonging to 900 languages. We filtered out languages with less than 100 phonological transcriptions available. The final dataset containing word-form phonological transcriptions included 731,145 word-forms from 192 languages. Information from each language was extracted using the `lingtypology` Python package (Moroz, 2017; Voronov & Forkel, 2019).

Data processing

We segmented word-form phonological transcriptions into syllabic candidates where possible. If available, we used explicit syllable boundary markers (`.`) or implicity syllable markers (`ˈ` for primary stress, `ˌ` for secondary stress). In the absence of syllabic markers, we assumed the word to be monosyllabic. We then classified the segments inside each resulting syllable candidate into consonants (`C`), vowels (`V`), and semi-vowels (`v`). Consecutive sequencies of vowels and semi-vowels (e.g., `V`, `VV`, `Vv`, `VVv`) where classified as vocalic clusters. Syllables with more than one vocalic cluster were flagged as multisyllabic, and were excluded from the dataset. We also remove duplicated syllables within each language (e.g., if the syllable /ka/ appeared in Spanish words /ˈka.sa/ and /kaˈpo/ in Spanish, it would appear more than once in the Spanish dataset), resulting in a syllabic repertoire for each language. We then assigned a sonority score to each segment in each syllable in a scale from 1 to 9, according to the `panphon` Python package (Mortensen et al., 2016). Finally, we evaluated whether each syllable followed or violated the SSP. Syllables with more than one peak (i.e., local minimum) in the sonority sequence were classified as SSP-violating, and as SSP-following otherwise. Syllables with a flat sonority contour were classifid as SSP-following.

Table 1: Examples of SSP-following and SSP-violating syllables in the dataset. Sonority index contours are shown for each syllable numerically and graphically.

Language	Syllable	Original word	Sonority index	
SSP-following				
Akkadian	/bar/	barbarum /ˈbar.ba.rum/	2-9-6	
Bats	/aɻ/	ⱔⱐ /aɻ/	9-7	
Western Kayah	/biɬ/	ꤘꤟꤢꤧ /biɬ.jaɬ/	2-8-1	
Silesian	/awz/	ausdruk /ˈawz.druk/	9-7-4	
Yoruba	/bɪ̀/	obinrin /ō.bɪ̀.ṛ̃ī/	2-8	
SSP-violating				
Bahnar	/hna:m/	hnam /hna:m/	6-5-9-5	
English	/bi:dz/	beads /bi:dz/	2-8-2-4	
Gothic	/do:gs/	𐌷𐌷𐍄𐌷𐌰𐌶𐍂𐍃 /ˈah.tɔ:.do:gs/	2-9-2-3	
Eastern Armenian	/fdor/	ֆտոր /fdor/	3-2-9-6	
Thai	/fɛ̄ksɻ/	แฟกซ์ /fɛ̄ksɻ/	3-9-1-3-1-1	

Data analysis

To model the prevalence of the SSP across languages, we aggregated, for each language in the dataset, the number of syllables that followed the SSP. This resulted in one data point per language. We used a multilevel, Bayesian logistic regression model to estimate the proportion of syllables that followed or violated the SSP across languages. This model estimated the average probability of the syllables of language family following the SSP. This model included each family language as a group with a random intercept. Equation 1 shows a detailed implementation of both models model. We implemented the model in the probabilistic programming language Stan (Carpenter et al., 2017) and CmdStanPy, its interface to its Python 3.12.7. We run eight Hamiltonian Monte Carlo chain of 2,000 iterations each (1,000 of which were warm-up iterations) to estimate the posterior distribution of the model.

Because we did not have access to the full syllabic repertoire of each language, available syllables from languages are the result of convenience sampling (e.g., in Wiktionary word entries are created by volunteers). The number of entries from each language, the amount of detail with which they are entered, and the type of words that are entered, varies systematically across languages. This might bias the estimates. For instance, if SSP-violating syllables are less frequent than SSP-following syllables in the lexicon of a language, and less frequent words are less likely to enter Wiktionary by volunteers, then the observed prevalence o fSSP for that language likely overestimates the prevalence of SSP-following languages. For this reason, this model included a term in the linear model γ/n_i aimed at capturing this potential overrepresentation of SSP-following syllables in datasets with fewer observations.

Likelihood:

$$
\begin{aligned}
y_j &\sim \text{Binomial } (n_j, \frac{\theta_j}{1-\theta_j}) \\
\theta_j &= \mu_j + u_j + (\gamma/n_j)
\end{aligned}
$$

where:

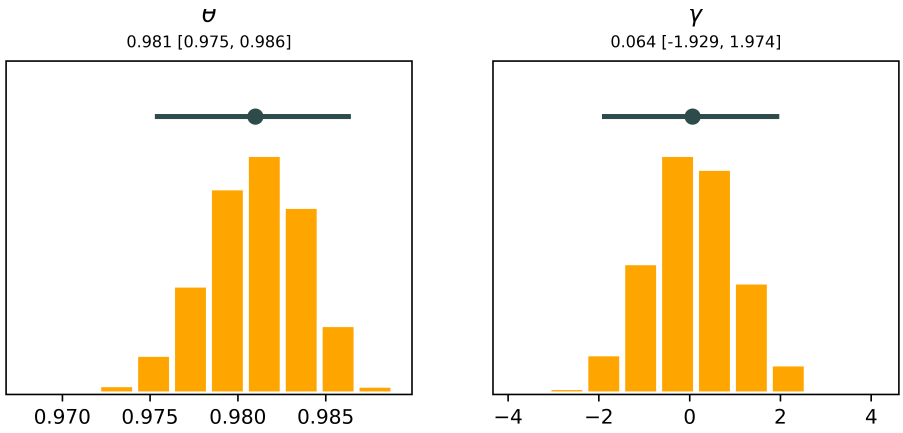
i represents each observation, corresponding to a language
y_j is the number of syllables in language i that follow SSP
u_j is a language-wise adjustment
n_i is the number of syllables total language i
γ introduces an adjustment to μ that is inverselly proportional to n_i

(1)

Prior:

$$
\begin{aligned}
\mu_j &\sim \text{Normal } (0, 5) \\
\sigma_u &\sim \text{Exponential } (4)
\end{aligned}
$$

Results



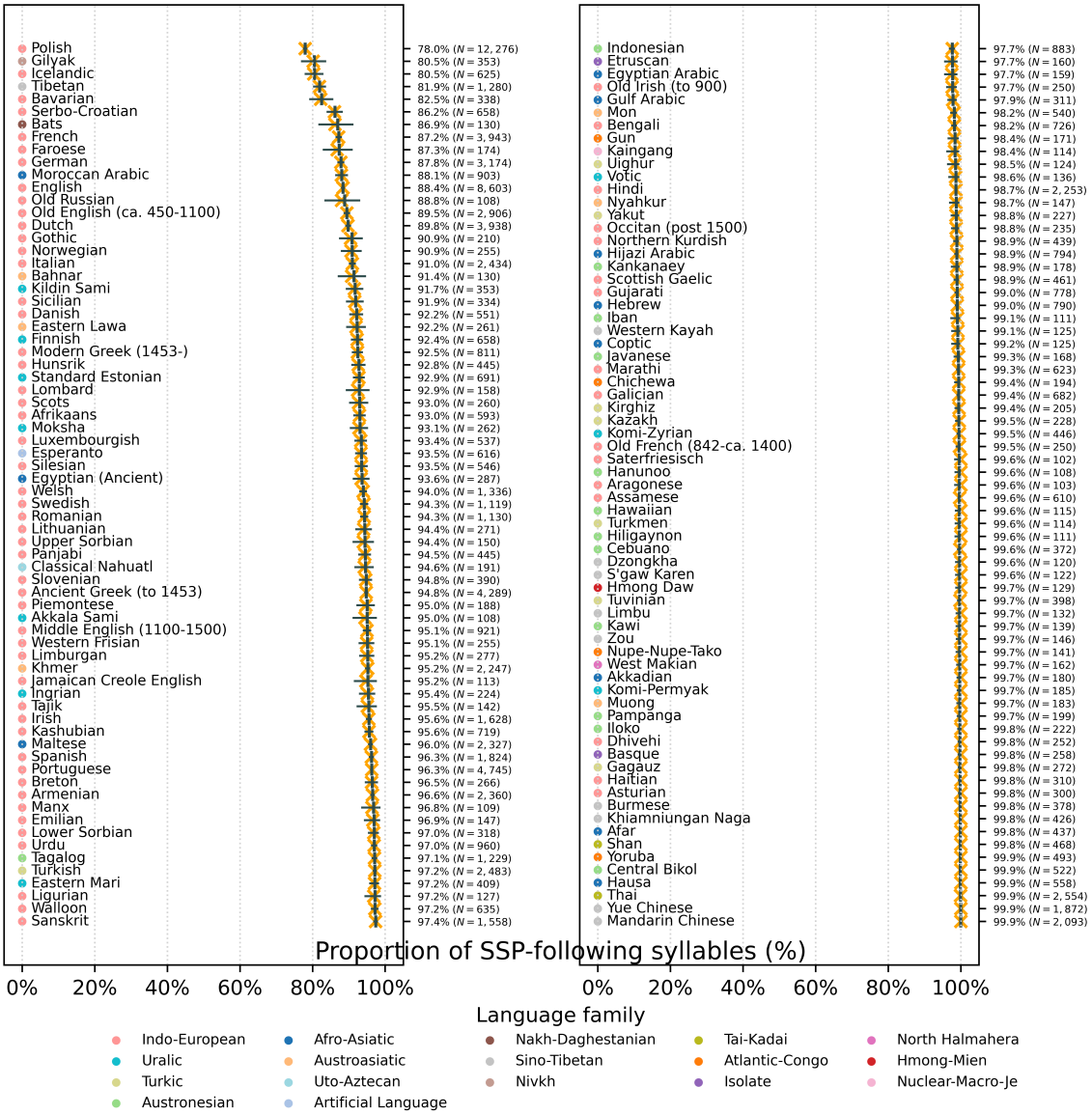


Figure 1: Posterior model-estimated proportion of distinct syllables that follow the Sonority Sequencing Principle (SSP) for each language family. The X-axis indicates the proportion of syllables that follow the SSP. The Y-axis indicates each language family. Black diamonds and intervals indicate the posterior mean proportion of SSP-following syllables for each language family, and its 95% highest density interval (HDI). Means and 95% HDI are also numerically reported next to each language family.

Analysis 2: Are SSP-following syllables more frequent than SSP-violating syllables?

1) Create a *syllabic frequency* measure that accounts for how often a syllable appears across words in each language, and that adjusts this prevalence by the lexical frequency of the words

it appears in (Frisch, 2015, for a similar approach). 1.1) For each language dataset, get the lexical frequency of each word in a lexical frequency database for that language. 1.2) *Syllabic frequency*: Add up the lexical frequency of the words each syllable appears in.

2) Fit a model (same basic stucture as before) that includes the *syllabic frequency* predictor, and compare its regression coefficient against zero (perhaps specifying a ROPE).

Method

Data collection

Extracting lexical frequencies

We extracted adult lexical frequencies for each word-form in the words dataset using the `wordfreq` Python package (Speer, 2022). This package provides an interface to retrieve lexical frequency data for words in over 40 languages. Lexical frequencies are extracted from a variety of sources according to their availability for each language, including Wikipedia, subtitles (e.g., OPUS Open subtitles 2018 and SUBTLEX), news, books, web text, Twitter, and Reddit, among others. At least three of these sources contribute to the calculation of lexical frequencies in each of the available languages. Importantly, lexical frequencies provided by `wordfreq` have have not been updated since September 2024 to prevent AI generated text from biasing the data. When lexical frequencies are not available for a particular language (e.g., the ISO 639-3 code does not match any available language in `wordfreq`), an equivalent language is used if available. Equivalent languages are searched according to their distance to the original language according to CLDR v38.1. This system assigns a numeric value to each pair of language codes based on whether both codes are assigned to identical languages with different names (e.g., Norwegian Bokmål → Norwegian, score 0), minor regional differences (score 1-3, e.g., Norwegian Bokmål → Norwegian), and languages with significant but unproblematic regional differences (score 10-24, e.g., Afrikaans → Dutch, Wu Chinese → Mandarin Chinese), etc. The rationale behind using equivalent languages to retrieve missing lexical frequencies is that users of close languages are likely to use similar words to refer to the same objects, and therefore lexical frequencies are expected to be equivalent across

Calculating syllabic frequencies

For each syllable, we aggregated the lexical frequency (frequency per million words) of each word the syllable was part of in the dataset. The rationale behind this step is that each time a word is uttered, each of its syllables is also uttered. Therefore, by calculating the lexical frequency of each word a given syllable is involved in, an estimation of syllabic frequency can be derived. Table 2 provides illustrative examples of how syllabic frequencies were calculated.

Table 2: Illustrative example of syllabic frequency calculation for syllables /ka/ and /ne/ across French and Spanish.

	French			Spanish		
	Word	Generic	CHILDES	Word	Generic	CHILDES
Syllable /ka/						
	/ka.do/	44.67	193.55	/kaˈreɾa/	125.89	21.31
	/ka.ʁa.mɛl/	3.80	8.53	/kaˈɾaxo/	41.69	0.44
	/ka.di/	1.12	9.75	/kaˈʃak/	0.93	0.44
	/ka.na.pe/	0.87	1.52	/kaˈjak/	0.93	0.44
	/kas.ka.dœʁ/	0.59	3.35	/kaˈt͡ʃaɾ/	0.19	0.44
PMW		51.05	216.71		169.64	23.08
Zipf		4.71	5.34		5.23	4.36
Syllable /ne/						
	/ne/	61.66	8.53	/neˈgoθjo/	85.11	1.78
	/ʒe.ne.ʁø/	13.18	0.30	/teɾmiˈne/	9.77	5.33
	/a.sɛ.zɔ.ne/	0.42	0.30	/neˈbeɾa/	4.27	11.10
	/a.se.zɔ.ne/	0.42	0.30	/neˈgɾita/	2.40	4.44
	/ʃi.fɔ.ne/	0.09	2.13	/neˈbado/	1.45	11.10
PMW		75.77	11.58		103.00	33.73
Zipf		4.88	4.06		5.01	4.53

Data analysis

We used a multilevel, Bayesian, distributional regression model to estimate the proportion of syllables that followed or violated the SSP across languages. The response variable (Frequency) was modelled using a log-normal distribution, with a location parameter μ and a scale parameter σ. We modelled both parameters using linear regression models, each with a global intercept (β_0, π_0) with a language-specific adjustment (u_{0j}, w_{0j}), a predictor variable SSP, which indexed whether the syllable followed the SSP (coded as `+0.5`) or violated the SSP (coded as `-0.5`) and its corresponding regression coefficient (β_1, π_1), with language-specific adjustments (u_{1j}, w_{1j}). Equation 2 provides a formal definition of the model. We specified weakly informative priors for the parrameters in the model. We run eight Hamiltonian Monte Carlo chain of 2,000 iterations each (1,000 of which were warm-up iterations) to estimate the posterior distribution of the model.

Likelihood:

$$
\begin{aligned}
y_{ij} &\sim \text{LogNormal}\ (\mu_{ij}, \sigma_{ij}) \\
\mu_{ij} &= \beta_0 + u_{0j} + (\beta_1 + u_{1j}) \cdot \text{SSP}_{ij} \\
\sigma_{ij} &= \exp(\pi_0 + w_{0j} + (\pi_1 + w_{1j}) \cdot \text{SSP}_{ij})
\end{aligned}
$$

Prior:

$$
\begin{aligned}
\beta_{[0,1]} &\sim \text{Normal}\ (0, 10) \\
\pi_{[0,1]} &\sim \text{Normal}\ (0, 5) \\
u_{[0,1]j} &\sim \text{Normal}\ (0, \tau_{uj}) \\
w_{[0,1]j} &\sim \text{Normal}\ (0, \tau_{wj}) \\
\tau_{uj}, \tau_{wj} &\sim \text{Exponential}\ (2)
\end{aligned}
\tag{2}
$$

Results

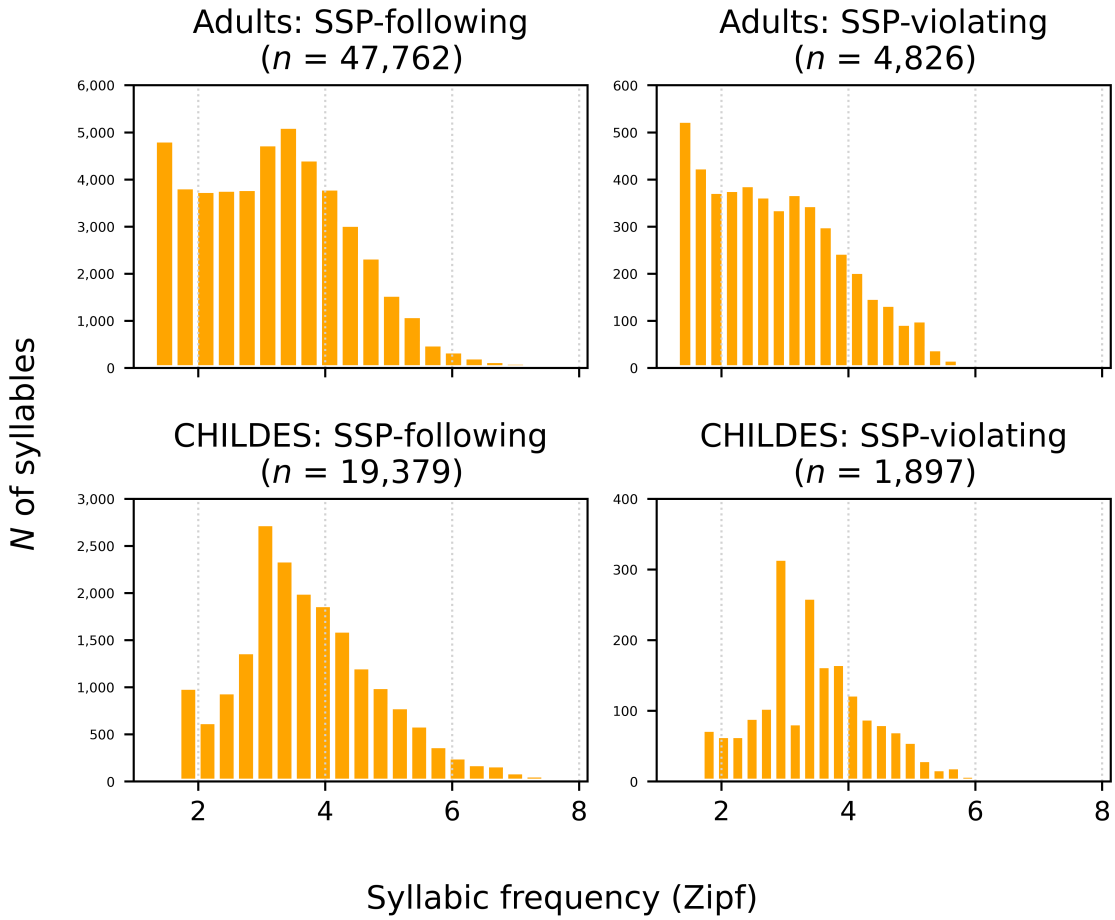
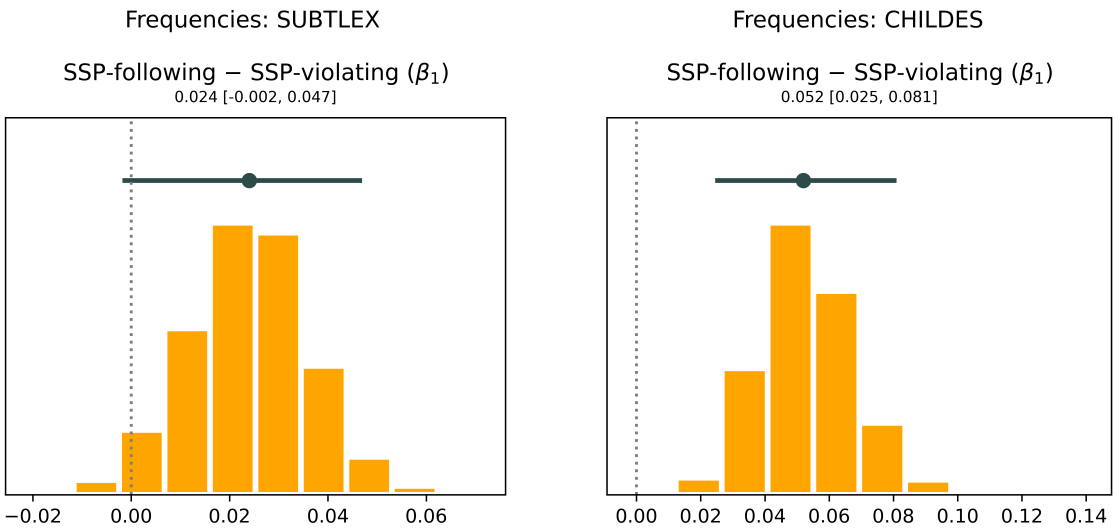


Figure 2: Distribution of syllabic frequencies across languages for SSP-followingg and SSP-violating syllables.



(a) Adult lexical frequencies (b) Infant lexical frequencies (CHILDES)

Figure 3: Posterior distribution of main parameters in Equation 2.

Appendix

Number of observations and SSP prevalence

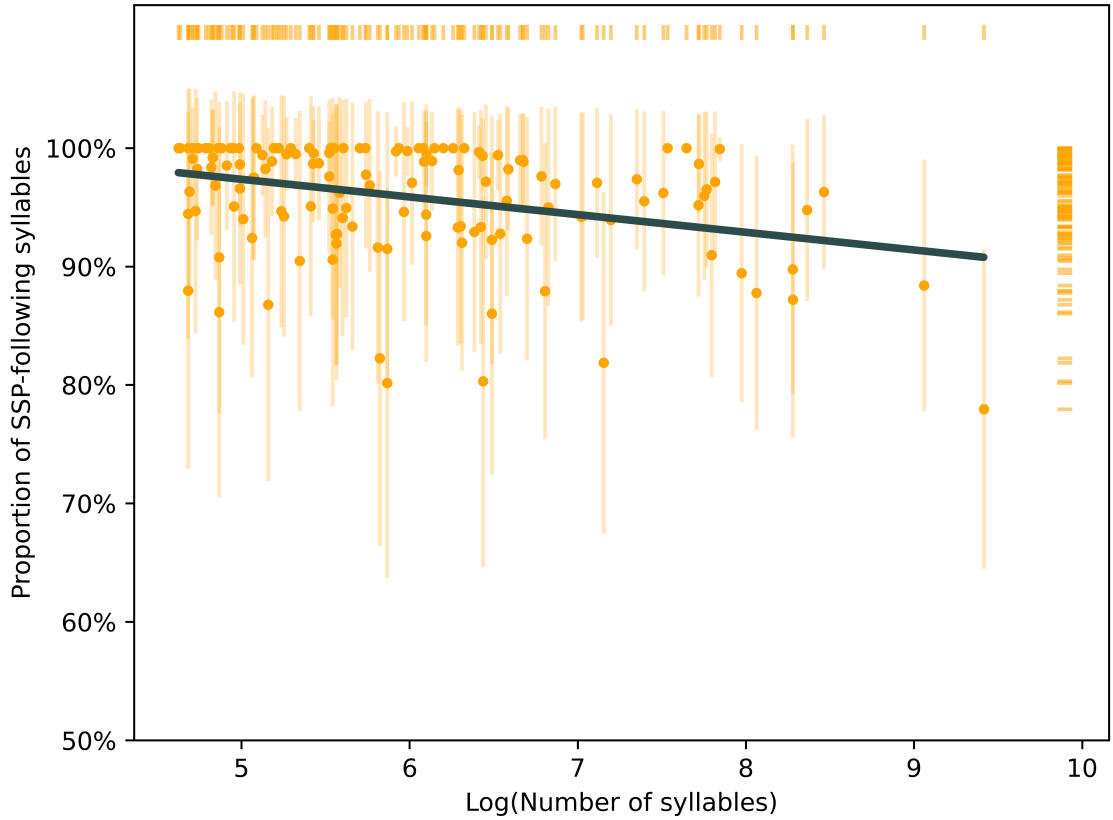


Figure 4: Relationship between the number of observed syllables and the SSP-prevalence in each language. The X-axis indicates the log-transformed number of observed syllables in the dataset. The Y-axis indicates the SSP prevalence (i.e., observed proportion of syllables that follow the SSP in the language's dataset). Each dot and interval indicate the data for one language, and the standard error of the SSP-prevalence. Marks in the top and right margin of the figure indicate the distribution of the observations. The predictions of a linear regression model between both variables are shown as a green line.

References

Bibliography

Albert, A., & Nicenboim, B. (2022). Modeling Sonority in Terms of Pitch Intelligibility With the Nucleus Attraction Principle. *Cognitive Science*, *46*(7), e13161. https://doi.org/10.1111/cogs.13161

Bijeljac-Babic, R., Bertoncini, J., & Mehler, J. (1993). How Do 4-Day-Old Infants Categorize Multisyllabic Utterances?. *Developmental Psychology*, *29*(4), 711.

Blevins, J. (1995). The Syllable in Phonological Theory. In J. Goldsmith (Ed.), *Handbook of Phonological Theory: Vol. 1. Handbook of Phonological Theory* (pp. 206–244). Blackwell.

Carpenter, B., Gelman, A., Hoffman, M. D., Lee, D., Goodrich, B., Betancourt, M., Brubaker, M., Guo, J., Li, P., & Riddell, A. (2017). Stan: A probabilistic programming language. *Journal of Statistical Software*, *76*, 1–32.

Clements, G. N. (1990). The Role of the Sonority Cycle in Core Syllabification. *Papers in Laboratory Phonology*, 283–333. https://doi.org/10.1017/cbo9780511627736.017

Fló, A., Brusini, P., Macagno, F., Nespor, M., Mehler, J., & Ferry, A. L. (2019). Newborns are sensitive to multiple cues for word segmentation in continuous speech. *Developmental Science*, *22*(4), e12802.

Frisch, S. A. (2015). A Preliminary Investigation of Quantitative Patterns in Sonority Sequencing. *Italian Journal of Linguistics*, *27*(1), 9–27.

Gomez, D., Berent, I., Benavides-Varela, S., Bion, R., Cattarossi, L., Nespor, M., & Mehler, J. (2014). Language Universals at Birth. *Proceedings of the National Academy of Sciences*, *111*(16), 5837–5841.

Halle, M., & Jones, L. G. (1971). *The Sound Pattern of Russian: A Linguistic and Acoustical Investigation*. Walter de Gruyter.

Itô, J. (1982). The Syllable Structure of Russian. *Unpublished Ms. University of Massachusetts-Amherst*.

Jany, C., Gordon, M., Nash, C. M., & Takara, N. (2007). How Universal Is the Sonority Hierarchy?: A Cross-Linguistic Acoustic Study. *Proceedings of the ICPhS*, 1401–1404.

Kawasaki-Fukumori, H., Ohala, J., Eliasson, S., & Jahr, E. (1997). Alternatives to the Sonority Hierarchy for Explaining Segmental Sequential Constraints. *Language and Its Ecology: Essays in Memory of Einar Haugen*, *100*, 343.

Lee, J. L., Ashby, L. F., Garza, M. E., Lee-Sikka, Y., Miller, S., Wong, A., McCarthy, A. D., & Gorman, K. (2020). Massively multilingual pronunciation modeling with WikiPron. *Proceedings of the Twelfth Language Resources and Evaluation Conference*, 4223–4228.

MacWhinney, B., & Snow, C. (1990). The child language data exchange system: An update. *Journal of Child Language*, *17*(2), 457–472.

Moroz, G. (2017). *lingtypology: easy mapping for Linguistic Typology*. https://cran.r-project.org/package=lingtypology

Mortensen, D. R., Littell, P., Bharadwaj, A., Goyal, K., Dyer, C., & Levin, L. (2016). Panphon: A resource for mapping IPA segments to articulatory feature vectors. *Proceedings of COLING 2016, The 26th International Conference on Computational Linguistics: Technical Papers*, 3475–3484.

Nordhoff, S., & Hammarström, H. (2011). Glottolog/Langdoc: Defining dialects, languages, and language families as collections of resources. In T. Kauppinen, L. C. Pouchard, & K. Keßler

(Eds.), *Proceedings of the First International Workshop on Linked Science 2011 (LISC2011), Bonn, Germany, October 24, 2011: Proceedings of the First International Workshop on Linked Science 2011 (LISC2011), Bonn, Germany, October 24, 2011* (pp. 1–7). CEUR Workshop Proceedings. http://iswc2011.semanticweb.org/fileadmin/iswc/Papers/Workshops/LISC/nordhoff.pdf

Parker, S. (2008). Sound Level Protrusions as Physical Correlates of Sonority. *Journal of Phonetics*, *36*(1), 55–90.

Parker, S. (2017). Sounding out Sonority. *Language and Linguistics Compass*, *11*(9), e12248. https://doi.org/10.1111/lnc3.12248

Prince, A., & Smolensky, P. (2004). Optimality in Grammar: Core Syllabification in Imdlawn Tashlhiyt Berber. In *Optimality Theory: Optimality Theory* (pp. 14–26). John Wiley & Sons, Ltd. https://doi.org/10.1002/9780470759400.ch2

Speer, R. (2022, September). *rspeer/wordfreq: v3.0*. Zenodo. https://doi.org/10.5281/zenodo.7199437

Voronov, M., & Forkel, R. (2019, June). *OneAdder/lingtypology: Release 0.8.5*. Zenodo. https://doi.org/10.5281/zenodo.3247007

Yin, R., Weijer, J. van de, & Round, E. R. (2023). Frequent violation of the sonority sequencing principle in hundreds of languages: how often and by which sequences?. *Linguistic Typology*, *27*(2), 381–403.